



JAVA OOPS Concepts

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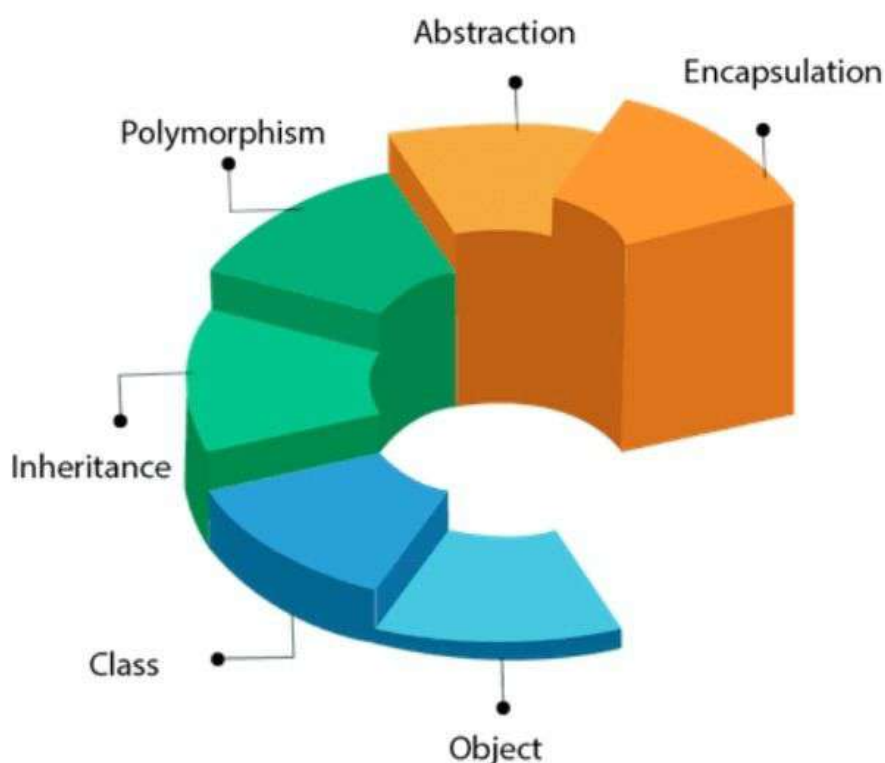
WHAT IS OOPS ?

What is OOPS concepts :

- **Object-Oriented Programming (OOP)** is a programming paradigm based on the concept of objects that contain data (fields) and behavior (methods). It focuses on designing software that closely represents real-world entities.

How OOPS is useful :

- Improves code reusability
- Enhances maintainability and scalability
- Makes programs easier to understand and manage
- Closely models real-world entities





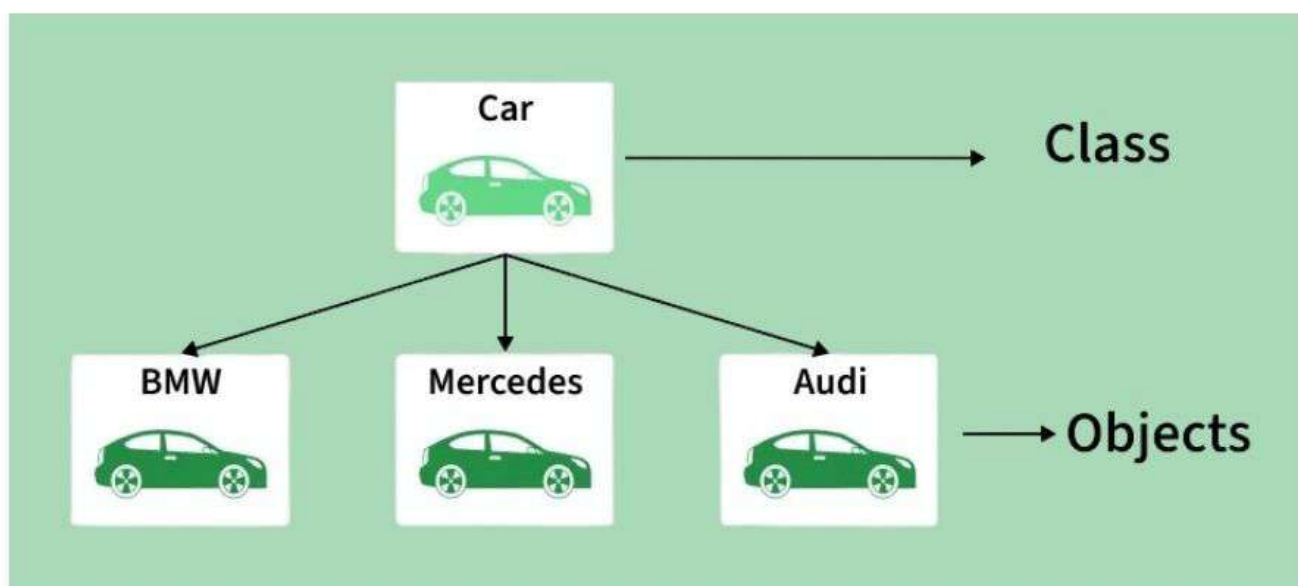
CLASS

What is Class ?

- A class is a user-defined blueprint or prototype used to create objects.
- It defines common properties (variables) and methods (behavior) for objects.
- Using a class, multiple objects can be created with the same structure and behavior, avoiding code repetition.

Class Declaration Components -

- Modifiers: public or default access
- Class Name: Starts with a capital letter
- Class Body: Enclosed within { }





OBJECT

What is Object ?

- An object is a basic unit of Object-Oriented Programming that represents a real-world entity.
- Objects interact with each other by invoking methods.
- They are responsible for executing the code and producing visible results to the user.

Main Components of an Object -

- State: Represents the object's attributes or data.
- Behavior: Represents the actions of the object through methods.
- Identity: Uniquely identifies an object.
- Method: A block of code that performs a specific task and returns a result.





ABSTRACTION

What is Abstraction ?

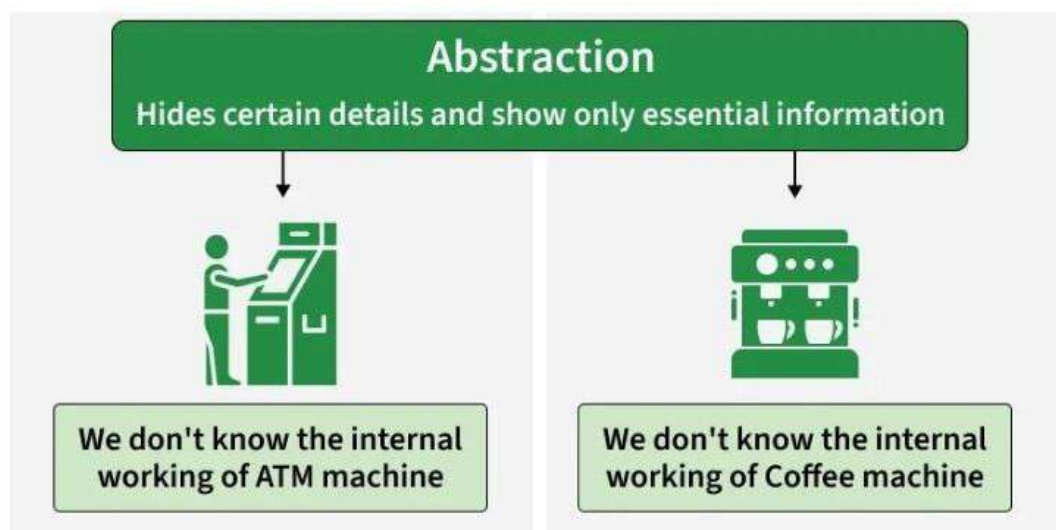
- **Abstraction is the process of hiding implementation details and showing only essential features. It helps users focus on what an object does, not how it does it.**

Advantages of Abstraction -

- **Hides complexity: Internal logic is hidden from the user.**
- **Improves maintainability: Changes in implementation don't affect user code.**
- **Enhances flexibility: Achieved using abstract classes and interfaces.**

Example -

- **ATM/Coffee Machine: User interacts with simple options, while internal working remains hidden.**





ENCAPSULATION

What is Encapsulation ?

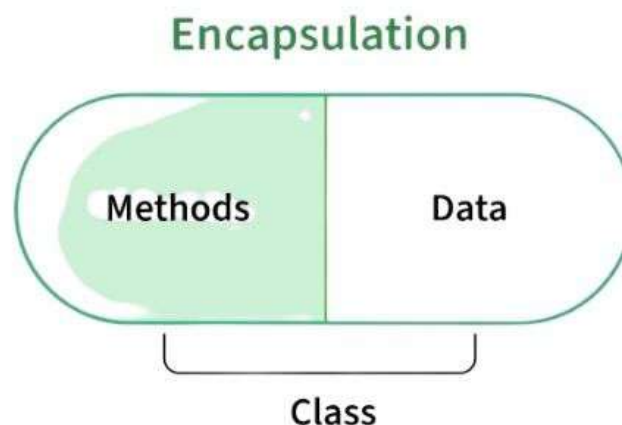
- Encapsulation means wrapping data and methods into a single unit (class).
- It restricts direct access to data from outside the class.

How Encapsulation is Achieved -

- Data members are declared private.
- Access is provided using public getter and setter methods.

Benefits of Encapsulation -

- Improves data security
- Enhances maintainability
- Provides controlled access to data





INHERITANCE

What is Inheritance ?

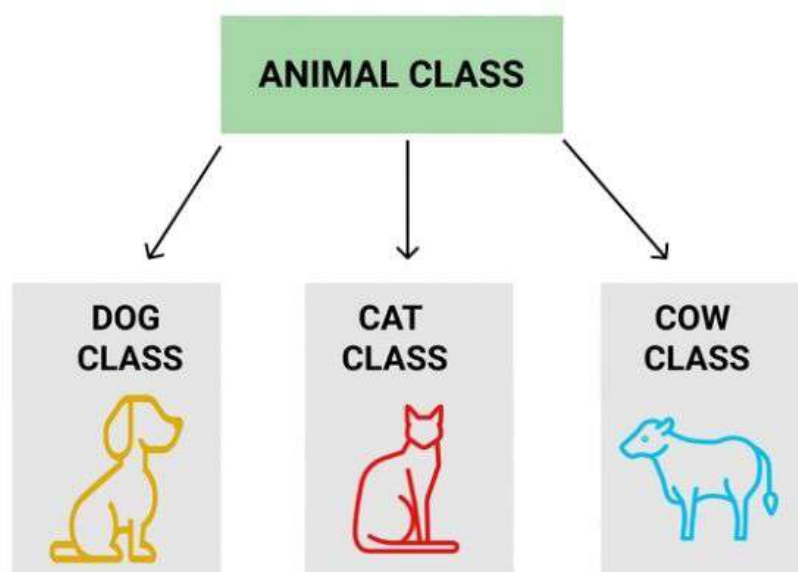
- Inheritance allows one class to acquire properties and methods of another class using the extends keyword.
- It represents an “is-a” relationship between classes.

Benefits -

- Code reusability
- Reduces redundancy

Key Points -

- Superclass: The class whose features are inherited.
- Subclass: The class that inherits the superclass.
- A subclass can reuse existing features and also add new ones.





POLYMORPHISM

What is Polymorphism ?

- Polymorphism means many forms (poly = many, morph = forms).
- It allows the same method or object to behave differently based on context.
- In Java, polymorphism mainly works at runtime based on the actual object type.

Types of Polymorphism -

- Compile-time Polymorphism (Method Overloading)
 - Same method name, different parameters
 - Resolved at compile time
- Runtime Polymorphism (Method Overriding)
 - Subclass provides its own implementation of a superclass method
 - Resolved at runtime based on object reference

